| INCH-POUND |

MIL-B-44395(GL)
14 December 1989

MILTITARY SPECIFICATION

BAG, PLASTIC (FOR HEATING WATER IN COMBAT VEHICLES)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a plastic bag to be used as a water heating vessel in combat vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-ES Natick MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8105

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SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder (21CFR PARTS 177 AND 182).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The bag shall be a two-layer laminated structure consisting of 0.002 inch (±0.0004 inch) thick, medium density polyethylene bonded to 0.00048 inch thick polyester. The component materials shall be food grade and free of foreign odors. The polyester shall comprise the outer layer of the bag.

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3.2 Style. The bag shall be flat style made from sheets or rolls and shall be clean and free of extraneous matter. The bag shall have two side seals and one bottom seal. The sides and bottom seals shall be formed by heat sealing and shall be 3/8 ± 1/8 inch wide. The bag shall be provided with an interlocking seal, bonded to the inner layer of the bag, not less than 1/2 inch and not more than 3/4 inch from the top of the bag. The interlocking seal shall be effected by applying pressure with the fingers to the seal area. The bag shall be provided with a 1/8 inch (+1/32, -1/16 inch) tear notch along the outside edge of one or both side seams. The tear-notch shall be located 1 ± 1/8 inch above the fill-line as specified in 3.4.

3.3 Dimensions. The inside dimensions of the bag shall be 7-1/4 ± 1/8 inch square.

3.4 Marking. The bag shall be marked with a horizontal line to indicate fill level. The fill-line shall be parallel to the bottom seal. The fill-line shall be located 2-3/8 ± 1/8 inch above the inside edge of the bottom seal. The ink used to print the fill-line shall be free of any carcinogenic agents.

3.5 Performance. The bag shall not leak when filled with 8 ounces of boiling water, and suspended with the interlocking seal down, 8 to 12 inches above a clean sheet for 10 minutes (See 4.2.2).

3.6 Federal Food, Drug, and Cosmetic Act. The bag shall conform to 21CFR, part 177 (except sections 177.1020, 177.1030, 177.1040, 177.1050 and 177.1480) and/or part 182.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed nesessary to ensure supplies and services confrom to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meeet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Material certification.

4.2.1.1 Purity. Purity of the material shall be accepted on the basis of the contractor's certification of conformance to the food grade requirements specified in 3.1 and 3.6.

4.2.1.2 Film thickness. The polyethylene and polyester shall be accepted on the basis of the contractor's certification of conformance to the thickness requirement specified in 3.1.

4.2.2 Performance. The bag shall be accepted on the basis of the contractor's certification of conformance to the leakage requirement specified in 3.5.

4.2.3 End item visual examination. The water heating bag shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one bag. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE I End item visual defects

Defect	Classification	
	Major	Minor
Tear, hole or split	101	
Foreign odor	102	
Inside dimensions not as specified	103	
Not heat sealed on three edges	104	
Fill-line not located as specified	105	
Heat seal width not as specified		201
Tear, nick, or notch missing or not located as specified		202
Dimension of tear-notch not as specified		203
Not clean		204
Interlocking closure not located as specified	106	
Interlocking closure not operable	107	

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4.2.4 Intermediate pack bag examination. The intermediate pack bags shall be examined for the defects listed in table II. The lot size shall be expressed in bags. The sample unit shall be one bag. The inspection level shall be S-3, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for one class of defects.

TABLE II. Intermediate pack bag and contents defects

Defect
Tear, hole, split in bag causing loss of contents
Interlocking closure missing
Labeling missing, incorrect, illegible or not as specified
Water heating bags not banded

4.4 Shipping container examination. Examination of filled and closed shipping containers shall be in accordance with the examination criteria in the appendix of PPP-B-636. In addition the following defects shall be added to the examination table contained therein.

Major - National Stock Number, item description, contract number, or date of pack marking missing, incorrect, or illegible
- More or less than 500 bags
- Box liner missing
Minor - Other required marking missing, incorrect, or illegible

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A preservation. One hundred water heating bags shall be banded together using a glued paper band. Each bundle of 100 banded bags shall then be placed in a intermediate flat polyethelene bag. The polyethylene bag shall be of sufficient size to contain the 100 water heating bags and shall be made of 0.003 inch thick polyethylene and have an interlocking closure seal which shall be closed after filling. The intermediate bag shall be marked as specified in 5.3.1.

5.2 Packing. Packing shall be level A or B as specified (see 6.2).

5.2.1 Level A packing. Five hundred water heating bags, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to non-metallic strapping for type II, class [illegible] loads.

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5.2.2 Level B Packing. Five hundred water heating bags, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, (Variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard shipping containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and unit loads shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each intermediate pack bag (see 5.1.1) shall be marked as follows:

100 water heating bags
Use to heat water in Heater, Ration, Individual: Mounted Troops.
Use purified water only.
Do not re-use.
Destroy if dirty or contaminated.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The bag is intended to be used as a vessel for heating water in the Heater, Ration, Individual, Mounted Troops (NSNs 7310-01-289-9645, 7310-01-304-8182, and/or 7310-01-304-867).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Level of packing required (see 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

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6.3 Subject term (key word) listing.

Hot water
Water bag
Hot beverage

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity

Army - GL

(Project 8105-0326)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44395(GL)	Bag,Plastic (For Heating Water in Combat Vehicles)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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